Google Ads API Tool: Design Document

x2 Ads Connector, an internal advertising tool for 47 Swag Co

Company	47 Swag Co, custom apparel, Long Island, New York
Website	https://47swagco.com
Contact	glendny@gmail.com
Tool type	Internal tool. Manages only 47 Swag Co's own Google Ads account(s), under our own manager account.
Users	The business owner only. No clients, agencies or third parties use the tool or can access it.
API access needed	Basic access

1. Purpose

47 Swag Co sells custom apparel (town and school spirit wear, embroidered hats and beanies, team and business apparel) through its online store to customers on Long Island. The tool lets the owner build and manage the company's Google Search campaigns and read their performance from one place, using an AI assistant (Claude) as the interface, instead of doing every step by hand in the Google Ads web interface.

2. How it works

```
Owner (in Claude chat) --> x2 Ads Connector (private Node.js server on Render.com) --> Google Ads API (REST)
                           secret URL, HTTPS only              OAuth 2.0 refresh token + developer token
```

- The connector is a small private web server (Node.js/TypeScript). It exposes a fixed list of actions ("tools") to Claude through the Model Context Protocol (MCP), and each tool calls the Google Ads REST API.
- Authentication: OAuth 2.0 with a refresh token for the owner's own Google account. Credentials are stored only as server environment variables and are never shown to the AI or to users.
- The endpoint is protected by a long secret in the URL; requests without it are rejected.

3. Features and API services used

Feature	Google Ads API services / resources
List accessible accounts	CustomerService.ListAccessibleCustomers
Reports: campaign performance, keyword performance and quality score, search terms, spend summaries	GoogleAdsService.Search (GAQL on campaign, keyword_view, search_term_view, customer)
Create a Search campaign with its own daily budget, Maximize Clicks bidding and Google Search network only (always created PAUSED)	CampaignBudgetService, CampaignService (via GoogleAdsService.Mutate)
Create ad groups; add keywords and negative keywords	AdGroupService, AdGroupCriterionService, CampaignCriterionService
Create responsive search ads (final URLs tagged with UTM parameters)	AdGroupAdService
Location targeting (counties such as Nassau and Suffolk County, NY, or a radius around a town)	CampaignCriterionService; geo_target_constant lookups via GoogleAdsService.Search
Pause / enable campaigns; change daily budgets	CampaignService, CampaignBudgetService

4. Safety controls

- **Dry run first:** every create action supports a dry-run mode that sends the request with *validate_only*, so Google checks it and nothing is created.
- **Explicit approval:** real creates, enabling a campaign and budget changes are refused unless the owner has approved the exact change (an explicit confirm flag).
- **Always paused:** new campaigns are always created PAUSED and only spend after the owner turns them on.
- **Budget cap:** a hard daily budget limit (default $30/day) is enforced in code. Larger budgets are refused.
- **Read-only switch:** a server setting turns off all changes.
- **Audit log:** every create call, including refused ones, is logged with time, account, action and result.
- **Atomic builds:** a full campaign is created in one all-or-nothing request, so it never leaves a half-built campaign behind.

5. Data handling and expected usage

- Report data is returned to the owner only and is not stored, sold or shared. No end-user personal data is collected.
- Expected volume is low: one to three accounts, a handful of campaigns, roughly 100-500 API operations per day. That is well within Basic access limits.
- The tool follows the Google Ads API Terms and Conditions and Required Minimum Functionality where applicable. Because it is an internal tool used only for our own account, it is not offered to other advertisers.